

Question 1 | Correct | Mark 1.00 out of 1.00

Given an array of N integer, we have to maximize the sum of $arr[i] * i$, where i is the index of the element ($i = 0, 1, 2, \dots, N$). Write an algorithm based on Greedy technique with a Complexity $O(n \log n)$.

Input Format:

First line specifies the number of elements- n

The next n lines contain the array elements.

Output Format:

Maximum Array Sum to be printed.

Sample Input:

5

2 5 3 4 0

Sample output:

40

Answer: (penalty regime: 0 %)

```

1  #include <stdio.h>
2  #include <stdlib.h>
3
4  int compare(const void *a, const void *b) {
5      long long diff = (long long)*(int *)a - (long long)*(int *)b;
6      if (diff < 0) return -1;
7      if (diff > 0) return 1;
8      return 0;
9  }
10
11 int main() {
12     int n;
13     if (scanf("%d", &n) != 1 || n <= 0) return 0;
14
15     int arr[n];
16     for (int i = 0; i < n; i++) {
17         scanf("%d", &arr[i]);
18     }
19
20     qsort(arr, n, sizeof(int), compare);
21
22     long long max_sum = 0;
23     for (int i = 0; i < n; i++) {
24         max_sum += (long long)arr[i] * i;
25     }
26
27     printf("%lld\n", max_sum);
28     return 0;
29 }
```

	Input	Expected	Got	
✓	5 2 5 3 4 0	40	40	✓

	Input	Expected	Got	
✓	10 2 2 2 4 4 3 3 5 5 5	191	191	✓
✓	2 45 3	45	45	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

[Back to Course](#)